

# Ardie Barry Sailis

## PhD Candidate in Pharmaceutical Sciences (Health)

Department of Pharmaceutical Life Sciences, Faculty of Pharmacy, Universiti Malaya

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Kuala Lumpur, Malaysia

### Profile

Toxicology researcher studying how e-cigarette use affects male reproductive health. My PhD at Universiti Malaya compares cigarette smokers, e-cigarette users and dual users in Malaysia, linking exposure to testosterone-related microRNAs, mitochondrial function and blood cell changes. I have published 14 peer-reviewed papers as sole or first author and review manuscripts for seven international journals. Open to roles in research, medical affairs, regulatory science and science communication.

### Education

**Doctor of Philosophy in Pharmacy (Health)** Dec 2023 – Dec 2026 (expected)  
Universiti Malaya, Kuala Lumpur  
Thesis on testosterone-related microRNAs, blood cell abnormalities and sexual desire in Malaysian cigarette smokers, e-cigarette users and dual users.

**Bachelor of Science in Microbiology, With Distinction** 2019 – 2023  
Universiti Malaya, Kuala Lumpur

### Experience

**Doctoral Researcher** Dec 2023 – present  
Department of Pharmaceutical Life Sciences, Faculty of Pharmacy, Universiti Malaya

- Conducted a comparative study of cigarette smokers, e-cigarette users and dual users in Malaysia.
- Analyzed testosterone-related microRNAs by qPCR alongside hormone and blood cell profiles.
- Used transmission electron microscopy to assess mitochondrial damage in e-cigarette users and controls.
- Wrote mechanistic and systematic reviews on e-cigarette toxicology, reproductive health and redox signaling.
- Developed an independent project, Cellular Signalling as Dynamic Regulatory Circuits, producing conceptual models of signal resolution in NRF2–KEAP1 redox signaling and YAP/TAZ mechanotransduction.

**Peer Reviewer** 2025 – present  
Seven international journals  
Reviews manuscripts in toxicology, pharmacology, reproductive health and public health for American Journal of Preventive Medicine, Journal of Hazardous Materials Advances, Pharmacological Reviews, Progress in Biophysics and Molecular Biology, Toxicology Reports, PLOS One and International Journal of General Medicine.

Supported laboratory operations during a full-time internship, covering microbiology procedures, water quality testing, sample handling, documentation and technical reporting under standard operating procedures.

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## Publications

- 14. YAP/TAZ as a mechanochemical signal-resolution circuit.**  
Ardie Barry Sailis  
*Progress in Biophysics and Molecular Biology*, 2026. [doi.org/10.1016/j.pbiomolbio.2026.101960](https://doi.org/10.1016/j.pbiomolbio.2026.101960)
- 13. Transcriptional condensates as kinetic filters for temporal control of gene expression.**  
Ardie Barry Sailis  
*Gene Reports*, 2026. [doi.org/10.1016/j.genrep.2026.102599](https://doi.org/10.1016/j.genrep.2026.102599)
- 12. Mitochondrial mechanisms in the potential male reproductive toxicity of e-cigarette-derived metal-containing nanoparticles.**  
Ardie Barry Sailis  
*Journal of Applied Toxicology*, 2026. [doi.org/10.1002/jat.70382](https://doi.org/10.1002/jat.70382)
- 11. Ferroptosis in e-cigarette aerosol-associated respiratory injury.**  
Ardie Barry Sailis  
*Archives of Toxicology*, 2026. [doi.org/10.1007/s00204-026-04486-w](https://doi.org/10.1007/s00204-026-04486-w)
- 10. Could e-cigarette devices generate inhalable micro- and nanoplastics? Exposure plausibility and reproductive relevance.**  
Ardie Barry Sailis  
*Toxicology Mechanisms and Methods*, 2026. [doi.org/10.1080/15376516.2026.2695155](https://doi.org/10.1080/15376516.2026.2695155)
- 9. Adipose as a Driver, Not a Bystander: A Modern Synthesis of Obesity-Related Erectile Dysfunction.**  
Ardie Barry Sailis, Muhamad Alfakri Mat Noh  
*Diabetes, Obesity and Metabolism*, 2026. [doi.org/10.1111/dom.70818](https://doi.org/10.1111/dom.70818)
- 8. E-cigarette aerosol constituents modulate Leydig cell steroidogenic pathways: Evidence from experimental models.**  
Ardie Barry Sailis, Muhamad Alfakri Mat Noh, Bey Fen Leo, Farid Nazer Faruqu, Anne Yee, Maw Shin Sim  
*Molecular and Cellular Endocrinology*, 2026. [doi.org/10.1016/j.mce.2026.112786](https://doi.org/10.1016/j.mce.2026.112786)
- 7. MicroRNA-mediated disruption of testosterone signaling associated with e-cigarette exposure.**  
Ardie Barry Sailis, Muhamad Alfakri Mat Noh, Bey Fen Leo, Farid Nazer Faruqu, Anne Yee, Maw Shin Sim  
*Environmental Toxicology and Pharmacology*, 2026. [doi.org/10.1016/j.etap.2026.104994](https://doi.org/10.1016/j.etap.2026.104994)
- 6. NRF2-KEAP1 as a redox signal-resolution circuit: Beyond the antioxidant switch.**  
Ardie Barry Sailis  
*Progress in Biophysics and Molecular Biology*, 2026. [doi.org/10.1016/j.pbiomolbio.2026.03.005](https://doi.org/10.1016/j.pbiomolbio.2026.03.005)
- 5. E-cigarettes and erectile dysfunction: biological mechanisms and research challenges.**  
Ardie Barry Sailis, Muhamad Alfakri Mat Noh, Bey Fen Leo, Farid Nazer Faruqu, Hui Yin Yow, Anne Yee, Maw Shin Sim  
*International Journal of Impotence Research*, 2026. [doi.org/10.1038/s41443-026-01300-0](https://doi.org/10.1038/s41443-026-01300-0)

4. **E-cigarette aerosols as systemic metabolic disruptors: integrated mitochondrial, circadian, and neurobehavioral mechanisms.**  
Ardie Barry Sailis  
*Toxicology Mechanisms and Methods*, 2026. [doi.org/10.1080/15376516.2026.2658739](https://doi.org/10.1080/15376516.2026.2658739)
  
3. **CYP1A1 as a conserved metabolic circuit linking environmental sensing to immune regulation.**  
Ardie Barry Sailis  
*Archives of Toxicology*, 2026. [doi.org/10.1007/s00204-026-04384-1](https://doi.org/10.1007/s00204-026-04384-1)
  
2. **Effects of secondhand exposure to e-cigarette aerosol on lung health: a systematic review.**  
Ardie Barry Sailis, Muhamad Alfakri Mat Noh, Bey Fen Leo, Farid Nazer Faruqu, Anne Yee, Maw Shin Sim  
*Journal of Public Health*, 2026. [doi.org/10.1007/s10389-026-02740-0](https://doi.org/10.1007/s10389-026-02740-0)
  
1. **Mitochondrial dysfunction induced by E-cigarettes.**  
Ardie Barry Sailis, Muhamad Alfakri Mat Noh, Bey Fen Leo, Farid Nazer Faruqu, Anne Yee, Maw Shin Sim  
*Toxicology*, 2025. [doi.org/10.1016/j.tox.2025.154339](https://doi.org/10.1016/j.tox.2025.154339)

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|-----------------------|-------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|
| Talks and conferences | PhD thesis seminar                                                                                                      |                                                                                                                                                             | 1 Oct 2026 |
|                       | Universiti Malaya                                                                                                       |                                                                                                                                                             |            |
|                       | Sexual Desire and Nicotine Dependence Among Smokers, E-Cigarette Users, and Dual Users in Malaysia: A Comparative Study |                                                                                                                                                             |            |
|                       | 2024<br>Poster presentation, ISAM Regional Meeting                                                                      |                                                                                                                                                             |            |
|                       | Early Career Researchers–Postgraduates Research Symposium                                                               |                                                                                                                                                             | 2024       |
|                       | ic-PHIR Conference                                                                                                      |                                                                                                                                                             | 2023       |
|                       | UiTM Puncak Alam                                                                                                        |                                                                                                                                                             |            |
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| Training              | Systematic Literature Review and Meta-Analysis Workshop                                                                 |                                                                                                                                                             | 2025       |
|                       | UV-Visible Double Beam Spectrophotometer Workshop, Shimadzu                                                             |                                                                                                                                                             | 2025       |
|                       | Quantitative PCR (qPCR) Workshop, Vazyme, Universiti Malaya                                                             |                                                                                                                                                             | 2024       |
|                       | ic-PHIR Metabolic Workshop, UiTM Puncak Alam                                                                            |                                                                                                                                                             | 2023       |
| Skills                | Laboratory                                                                                                              | qPCR, transmission electron microscopy, molecular biology techniques                                                                                        |            |
|                       | Data                                                                                                                    | SPSS, bioinformatics, data interpretation                                                                                                                   |            |
|                       | Evidence                                                                                                                | Systematic reviews, literature synthesis, mechanistic toxicology                                                                                            |            |
|                       | Writing                                                                                                                 | Scientific writing, manuscript preparation, peer review, research conceptualization                                                                         |            |
|                       | Research areas                                                                                                          | E-cigarette and tobacco toxicology, reproductive health, pharmacology, toxicogenomics, molecular systems biology, mechanotransduction and YAP/TAZ signaling |            |
| Languages             | Malay                                                                                                                   |                                                                                                                                                             |            |
|                       | Native or bilingual proficiency                                                                                         |                                                                                                                                                             |            |

**References**

Available on request.